

Verifiable-Reward Reinforcement Learning for Post-OCR Transliteration of Kazakh Arabic Script (Töte Jazu): A Feasibility Study

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Abstract—Töte jazu, the Arabic-script orthography historically used to write Kazakh, is a low-resource target for optical character recognition (OCR): no real-scan benchmark exists, and the script’s abjad nature creates systematic many-to-one ambiguities that a purely visual model cannot resolve. We study *post-OCR transliteration*—mapping recognized Arabic-script characters to Cyrillic—as a sequence-level task amenable to reinforcement learning with a *verifiable reward* (RLVR). We implement Group Relative Policy Optimization (GRPO) with a deterministic edit-distance verifier, a KL penalty to a supervised reference policy, and a short supervised warm-start, all trainable on CPU. On a controlled synthetic testbed with linguistically motivated homographs, GRPO reduces held-out character error rate (CER) of a lightly supervised base policy from 0.605 to 0.418 (31% relative) using only the reward signal. We report two strong baselines—a deterministic lookup table (0.297 CER) and a fully supervised model (0.277 CER)—that GRPO does *not* surpass on this near-deterministic cipher, and we argue this is the expected outcome: RLVR’s value is concentrated in the ambiguous, noisy regime of real data, which we identify as the next stage. This is a work-in-progress feasibility study; all results are reproducible on CPU.

Index Terms—low-resource OCR, transliteration, reinforcement learning, verifiable reward, GRPO, Kazakh, Arabic script

I. INTRODUCTION

Kazakh has been written in three scripts over the last century; the Arabic-based *Töte jazu* remains in use among diaspora communities and in a large body of pre-Cyrillic printed material. Digitizing this material requires OCR, but the language is low-resource for this task in two compounding ways. First, to our knowledge no public benchmark of *real scanned* Töte jazu pages with ground-truth transcriptions exists; prior Kazakh Arabic-script OCR work relies on synthetic renderings. Second, the script is an *abjad*: short vowels are under-specified and several Cyrillic phonemes collapse onto a single Arabic-script form, so that a visually perfect recognizer still faces an inherently ambiguous inverse mapping back to Cyrillic.

This motivates treating transliteration as a distinct, learnable stage placed *after* the visual recognizer, and optimizing it for a sequence-level metric (CER) rather than per-token likelihood. Reinforcement learning with a verifiable reward (RLVR), in

which a deterministic checker scores each sampled output and no learned reward model is used, is a natural fit: the correctness of a transliteration is checkable by string comparison, exactly as a math answer is checkable in recent RLVR work [?].

This paper is a feasibility study with three contributions. (i) We give a CPU-trainable RLVR pipeline for post-OCR transliteration based on GRPO [?] with an edit-distance verifier and a KL anchor to a supervised reference. (ii) On a controlled synthetic testbed we show that GRPO improves a lightly supervised base policy using the reward signal alone. (iii) We report—rather than omit—two baselines that GRPO does not beat on this testbed, and we explain why, delimiting precisely the regime in which RLVR is expected to help and motivating the move to real data and a real OCR front-end.

II. RELATED WORK

Transformer OCR. TrOCR [?] couples a vision-transformer encoder with a text-transformer decoder and is the standard transfer-learning base for low-resource scripts; the small variant is trainable on a single commodity GPU. **Attention seq2seq.** Our transliteration policy is a GRU encoder-decoder with dot-product attention [?], which lets the decoder align output character i to the relevant input position and is what allows generalization beyond memorized training words. **Verifiable-reward RL.** Policy-gradient methods [?], [?] and RL from human feedback [?] established the optimization machinery; GRPO [?] removes the value network by using group-relative advantages and is the method we adopt. The defining feature of our setup is that the reward is a *deterministic verifier*, not a learned model.

III. METHOD

A. Target pipeline

The intended system has three stages: (1) synthetic image rendering of Cyrillic Kazakh text through a script-conversion cipher and Arabic-script fonts, with augmentation, to bootstrap training data; (2) a fine-tuned TrOCR recognizer producing Arabic-script text; and (3) an RLVR *post-correction / transliteration* module producing Cyrillic. Stages (1)–(2) are future work (Section ??); this paper isolates and validates stage (3).

B. Policy and supervised warm-start

The policy π_θ is a bidirectional-GRU encoder and a GRU decoder with dot-product attention (embedding 64, hidden 128). Pure RL from a random initialization fails to leave the cold-start region, so we apply a short supervised fine-tuning (SFT) warm-start with teacher forcing before RL, matching the SFT→RL recipe of modern alignment pipelines.

C. GRPO with a verifiable reward

For a source string we sample a group of $G = 6$ candidate transliterations. Each candidate y is scored by a deterministic verifier

$$r(y, y^*) = (1 - \text{CER}(y, y^*)) + 0.5\mathbb{I}[y = y^*], \quad (1)$$

where CER is the normalized Levenshtein distance and y^* is the gold Cyrillic. Group-relative advantages $\hat{A}_i = (r_i - \bar{r})/(\sigma_r + \varepsilon)$ replace a learned value function [?]. The objective combines the policy-gradient term with a KL penalty to the frozen SFT reference π_{ref} ,

$$\mathcal{L} = -\mathbb{E}[\hat{A} \log \pi_\theta(y)] + \beta \widehat{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}), \quad (2)$$

with $\beta = 0.04$ and the KL estimated on sampled tokens as $\log \pi_\theta(y) - \log \pi_{\text{ref}}(y)$. The KL term is essential: without it, RL applied to an already-converged policy drifts and *degrades* held-out accuracy. Sampling temperature is annealed from 1.1 to 0.6. The entire procedure runs on CPU.

IV. EXPERIMENTAL SETUP

A. Controlled testbed

Because no real aligned corpus is yet integrated, we evaluate on a synthetic cipher that converts Cyrillic to a pseudo-Arabic alphabet. Crucially, the cipher is *not* a clean bijection: five linguistically motivated letter pairs (front/back vowel and semivowel pairs that genuinely share one Arabic-script form, e.g. the i/y and the close-vowel pairs) are mapped to a single glyph, so the inverse task carries local, context-dependent ambiguity that mirrors real Töte jaz. We stress that this is a *machinery testbed*, not a linguistic claim. The corpus is 420 synthetic words, split 336/84 into train/test; source and target vocabularies are 29 and 40 symbols.

B. Configurations

We compare four systems on the same held-out set: (a) a deterministic *lookup table* mapping each Arabic glyph to its most frequent Cyrillic pre-image in the training split; (b) *SFT (light)*, 900 warm-start steps, used as the RL base policy; (c) *SFT→GRPO*, the light base refined by 1500 GRPO steps; and (d) *SFT (full)*, 2500 supervised steps. During RL we monitor a 30-word subset; the final table reports the full 84-word held-out set.

TABLE I

HELD-OUT RESULTS ON THE CONTROLLED TESTBED (FULL 84-WORD TEST SET). LOWER CER AND HIGHER EXACT-MATCH ARE BETTER.

Method	Exact-match $\uparrow$	CER $\downarrow$
Deterministic lookup table	0.143	0.297
SFT (light, 900) — RL base	0.036	0.605
SFT→GRPO (RLVR)	0.131	0.418
SFT (full, 2500)	0.286	0.277

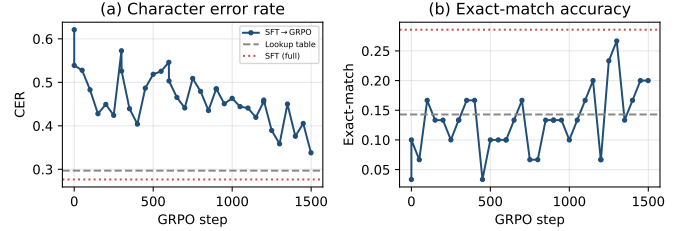


Fig. 1. GRPO learning curves on the held-out monitoring subset ($n=30$). Dashed: deterministic lookup table. Dotted: fully supervised model. RLVR steadily reduces CER over its supervised base but does not reach the stronger baselines on this near-deterministic cipher.

V. RESULTS

Table ?? and Fig. ?? report the outcome. Using only the verifiable reward and a KL anchor, GRPO improves the light SFT base from 0.605 to 0.418 CER (31% relative) and from 0.036 to 0.131 exact-match. This confirms that the RLVR machinery learns a real signal on CPU.

We deliberately report the two baselines GRPO does *not* beat. The deterministic lookup table reaches 0.297 CER, and full supervised training 0.277. The narrow gap between these two is itself informative: even unlimited supervision barely improves on a trivial table, because the synthetic words have *no morphological context* with which to disambiguate the shared-glyph pairs. On a near-deterministic mapping with no exploitable context, a likelihood-trained or rule-based model is the right tool, and RLVR’s exploration only adds variance. This is the expected boundary of the method, not a contradiction of it: RLVR is predicted to help precisely where the target is ambiguous and context-resolvable and where the input is noisy—neither of which the present testbed provides.

VI. LIMITATIONS AND FUTURE WORK

The dominant limitation is that the corpus is a synthetic cipher rather than real orthography, and there is as yet *no OCR front-end*: the system transliterates clean symbol strings, not recognizer output. Three concrete next stages follow directly. First, replace the cipher with a real aligned corpus—an Arabic-script / Cyrillic parallel text—so that ambiguity and morphological context are authentic; the RL code is unchanged and accepts such pairs directly. Second, fine-tune a small TrOCR model on synthetically rendered pages (feasible on a free cloud T4) to produce the realistic, *error-bearing* input that makes post-correction non-trivial. Third, evaluate RLVR specifically on the disambiguation and OCR-error cases

against the lookup-table and SFT baselines, where the method is expected to show its advantage. We also plan an ablation of the KL coefficient β and group size G .

VII. CONCLUSION

We presented a CPU-trainable, fully reproducible RLVR pipeline (GRPO with a deterministic verifier and a KL-anchored reference) for post-OCR transliteration of Kazakh Arabic script. On a controlled testbed the method demonstrably improves a supervised base from the reward signal alone, while honest baselines locate the regime—ambiguous, context-rich, noisy real data—in which it is expected to pay off. The contribution is an end-to-end, verifiable scaffold and an honest map of where the research must go next, rather than a premature state-of-the-art claim on a toy task.

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